

Homework 1

MATH 346/377 - Winter 2025

Due on Sunday, January 26, 11:59pm on Crowdmark

Total of points = 80 points

A complex number $\alpha \in \mathbb{C}$ is an *algebraic number* if there exists a polynomial $f \in \mathbb{Q}[x]$ such that $f(\alpha) = 0$. A *minimal polynomial* of α is a polynomial in $\mathbb{Q}[x]$ of minimal degree that has α as a root. Such a polynomial is not unique (you can always multiply by an element in $\mathbb{Q}^\times$ to get another minimal polynomial). In the textbook and in the first lecture notes, to get a unique minimal polynomial we required that its coefficients are coprime integers and that the leading coefficient is positive. Sometimes, it is also convenient to replace¹ this condition by the condition that the minimal polynomial is *monic* (in that case the coefficients are a priori only rational numbers, not integers). An algebraic number $\alpha \in \mathbb{C}$ is an algebraic integer if it has a monic minimal polynomial with integer coefficients.

For the following exercise you might need the following Lemma, also known as Gauss' Lemma. It says that if an integral polynomial can be factored over $\mathbb{Q}[x]$, it can be factored over $\mathbb{Z}[x]$.

Lemma 0.1. *Let $f(x) \in \mathbb{Z}[x]$ be a polynomial with integer coefficients and suppose that we can factor $f(x) = A(x)B(x)$ with $A(x), B(x) \in \mathbb{Q}[x]$ two polynomials. We can find $r, s \in \mathbb{Q}$ such that $a(x) = rA(x)$ and $b(x) = sB(x)$ are in $\mathbb{Z}[x]$ and $f(x) = a(x)b(x)$.*

1. (14 points) Show the following.
 - (a) (5 points) Let $m(x) \in \mathbb{Q}[x]$ be a minimal polynomial of an algebraic number $\alpha \in \mathbb{C}$. If $f(x) \in \mathbb{Q}[x]$ has α as a root, then $m(x)$ divides $f(x)$.
 - (b) (3 points) Deduce that an algebraic number α has a unique monic minimal polynomial.
 - (c) (6 points) An algebraic number $\alpha \in \mathbb{C}$ is an algebraic integer if and only if α is the root of *some* monic polynomial with integer coefficients. (So not necessarily the minimal one).

¹Given a minimal polynomial that has coprime integer coefficients but is not monic, one obtains a monic minimal polynomial with rational coefficients by dividing by the leading coefficient. Conversely, given a monic minimal polynomial with rational coefficients, we get a minimal polynomial with coprime integer coefficients by clearing the denominators. For example $2x^2 - 1$ and $x^2 - 1/2$ are both minimal polynomials of $1/\sqrt{2}$. The first one has coprime integer coefficients, the second one has rational coefficients but is monic.

Solution:

- (a) Let $m(x) \in \mathbb{Q}[x]$ be a minimal polynomial of degree $d \geq 1$. Let $f(x) \in \mathbb{Q}[x]$ of degree $d' \geq 1$ such that $f(\alpha) = 0$. By the minimality property, we must have $d \leq d'$, otherwise m would not be a minimal polynomial. By polynomial division, we can write

$$f(x) = qm(x) + r(x) \quad (1)$$

with $q(x), r(x) \in \mathbb{Q}[x]$ and $0 \leq \deg(r) \leq d - 1$. If we plug-in α on both sides, we find that $0 = r(\alpha)$. The polynomial r must have degree 0, otherwise this would contradict the minimality of m . Hence, we have $r = 0$ and $f(x)(x) = qm(x)$. Since $m(x)$ and $m'(x)$ are both monic, we have $q = 1$.

- (b) Let $m(x) \in \mathbb{Q}[x]$ be a monic minimal polynomial of degree $d \geq 1$. Suppose that there is another monic minimal polynomial $m'(x) \in \mathbb{Q}[x]$ of degree $d' \geq 1$. By part a), we must have that m divides m' . but also m' divides m . Hence, we have $m'(x) = qm(x)$. Since $m(x)$ and $m'(x)$ are both monic, we have $q = 1$.
- (c) One direction is immediate. For the other direction, suppose that α is the root of some monic polynomial $f(x) \in \mathbb{Z}[x]$. Let $M(x) \in \mathbb{Q}[x]$ be a monic minimal polynomial of α . By part a), we have $f(x) = G(x)M(x)$. By the Lemma, we can write $f(x) = g(x)m(x)$ where $g(x) = rG(x)$ and $m(x) = sM(x)$ both have integer coefficients. Note that $m(x)$ is still a minimal polynomial of α . Since $f(x)$ is monic, the polynomials $g(x)$ and $m(x)$ must have leading coefficients ± 1 . By replacing r, s by $-r, -s$ if necessary, we can make sure that the leading coefficient of $m(x)$ is 1 *i.e.* that $m(x)$ is monic. It follows that α has a monic minimal polynomial with integer coefficients, hence α is an algebraic integer.

2. (13 points) The goal of this problem is to show that the set of algebraic integers form a ring. We say that a ring R is finitely generated over $\mathbb{Z}$ if there exist elements $r_1, \dots, r_k \in R$ such that every element $r \in R$ can be written

$$r = n_1 r_1 + \dots + n_k r_k \quad (2)$$

for some integers $n_1, \dots, n_k$ in $\mathbb{Z}$.

- (a) (0 points) If $R \subset \mathbb{C}$ is finitely generated and $\alpha \in R$, then α is an algebraic integer. *This part is trickier and you can use it without proof in the rest of the exercise. This part will not be marked but if you want to prove it, the idea is to represent α by a matrix and to use Cayley-Hamilton theorem.*

(b) (6 points) Show that if $\alpha \in \mathbb{C}$ is an algebraic integer, then

$$\mathbb{Z}[\alpha] = \{f(\alpha) \mid f \in \mathbb{Z}[x]\} = \left\{ \sum_{i=0}^d a_i \alpha^i \mid a_i \in \mathbb{Z} \right\} \quad (3)$$

is finitely generated.

- (c) (4 points) Use the same idea to show that $\mathbb{Z}[\alpha, \beta] = (\mathbb{Z}[\alpha])[\beta]$ is finitely generated if $\alpha, \beta \in \mathbb{C}$ are algebraic integers.
- (d) (3 points) Deduce that $\alpha + \beta$ and $\alpha\beta$ are algebraic integers.

Solution:

(a)

- (b) Since α is an algebraic integer, there is a monic polynomial $g \in \mathbb{Z}[x]$ of degree n such that $g(\alpha) = 0$. Let us prove that we can take as basis $1, \alpha, \dots, \alpha^{n-1}$. For that, it is enough to show that any power α^k with $k \geq n$ can be written as a linear combination of $1, \alpha, \dots, \alpha^{n-1}$ with integers coefficients. We write

$$g(x) = x^n + h(x) \quad (4)$$

where $h(x) \in \mathbb{Z}[x]$ has degree $\deg(h) \leq n-1$. Since $g(x)$ is monic, for any $k \geq n$ we can perform polynomial division of x^k by $g(x)$ over $\mathbb{Z}[x]$ and write

$$x^n = q(x)g(x) + r(x) \quad (5)$$

with $q(x), r(x) \in \mathbb{Z}[x]$ and $\deg(r) \leq n-1$. If we evaluate at α we find that $\alpha^k = r(\alpha) \in \mathbb{Z}[\alpha]$ is a linear combination of $1, \alpha, \dots, \alpha^{n-1}$.

- (c) The idea is similar. Suppose that α and β are roots of monic polynomials $f, g \in \mathbb{Z}[x]$ of degree n and m respectively. Let us show that a basis is given by all $\alpha^i \beta^j$ for $0 \leq i \leq n-1$ and $0 \leq j \leq m-1$. An element in $(\mathbb{Z}[\alpha])[\beta]$ is a polynomial in β with coefficients in $\mathbb{Z}[\alpha]$:

$$p_0(\alpha) + p_1(\alpha)\beta + \dots + p_d(\alpha)\beta^d, \quad p_i \in \mathbb{Z}[\alpha]. \quad (6)$$

We have already shown in part a) that each $p_i(\alpha)$ can be written as a linear combination of $1, \alpha, \dots, \alpha^{n-1}$. The same proof shows that each β^k for $k \geq m$ can be written as a linear combination of $1, \beta, \dots, \beta^{m-1}$.

- (d) Since $\alpha + \beta$ and $\alpha\beta$ are both in the finitely generated ring $\mathbb{Z}[\alpha, \beta]$, it follows from part a) that they are algebraic integers.

Pari GP is open-source computer algebra system designed for fast computations in number theory. You can run it on your browser at the following link <https://pari.math.u-bordeaux.fr/gpasm.html>.

3. (19 points) The command `algdep(x,d)` takes as input a number x (computed to some accuracy), an integer d and tries to find a polynomial of degree d whose coefficients are as small as possible and for which x is a root.

- (a) (2 points) The accuracy is determined by the command `\p`. Display π with an accuracy of 5 digits: first type `\p5` followed by `Pi`. Vary the precision to get a sense of how it works. Set the precision to 5 and use the command `algdep(Pi,6)`. What do you observe and what does it suggest? Change the precision to 500 digits and run the command again.

The goal of this question is to show the limitations of numerical computations. If the accuracy is very high compared to the degree, the command `algdep(x,d)` can give strong evidence that a number is algebraic, but it does not prove that it actually is.

- (b) (4 points) From now on set the accuracy to 500 digits. By Exercise 2 the real number

$$x = \sqrt{2} + \sqrt{3} \quad (7)$$

is an algebraic integer. Run the command `algdep(x,d)` and increase the input d . Find a possible minimal polynomial of x , and verify by hand that x is indeed a root.

Remark: the command for $\sqrt{2}$ is `sqrt(2)`.

- (c) (4 points) Show that if $\alpha \in \mathbb{C}$ is a nonzero algebraic number, then $\frac{1}{\alpha}$ is an algebraic number. Use Pari GP to find a minimal polynomial of

$$x = \frac{1}{\sqrt{2} + \sqrt{3}}. \quad (8)$$

What do you observe? Write down an example of an algebraic integer α such that $1/\alpha$ is *not* an algebraic integer.

- (d) (4 points) Show that if a real number α is an algebraic integer, then $\sqrt{\alpha}$ is also an algebraic integer. Deduce from Exercise 2 that

$$x = \sqrt{\sqrt{3} + \sqrt{2}} + \sqrt{\sqrt{3} - \sqrt{2}} + 2i. \quad (9)$$

is an algebraic integer and find a minimal polynomial of x using Pari GP (you do not need to verify your result by hand).

Remark: the command for $a + ib$ is `a+b*I`.

- (e) (5 points) Let $t \in \mathbb{Q}$ be a rational number. Show that $2\cos(\pi t)$ is an algebraic integer. Use Pari GP to find a possible minimal polynomial of $\cos(3\pi/5)$.

Remark: the command is `cos(Pi*3/5)`. You don't have to verify that $\cos(3\pi/5)$ is indeed a root but you can try!

Solution:

- (a) It suggests that π is algebraic, which is wrong.
- (b) Pari GP suggests the minimal polynomial $x^4 - 10x^2 + 1$. A direct computation shows that $\sqrt{2} + \sqrt{3}$ is indeed a root of this polynomial.
- (c) If α is an algebraic number then there is a polynomial f such that

$$f(\alpha) = c_0 + c_1\alpha + \cdots + c_n\alpha^n = 0. \quad (10)$$

We find that

$$\alpha^n f(1/\alpha) = c_0\alpha^n + c_1\alpha^{n-1} + \cdots + c_n = 0, \quad (11)$$

hence $1/\alpha$ is a root of the polynomial $g(x) = x^n f(1/x)$ (obtained by reversing the coefficient). For the inverse of $\sqrt{2} + \sqrt{3}$, we see from the previous proof (or from Pari GP) that it has the same minimal polynomial as $\sqrt{2} + \sqrt{3}$, hence it is also an algebraic integer.

But in general the inverse of an algebraic integer is not an algebraic integer. For example $\alpha = \sqrt{2}$ is an algebraic integer since its minimal polynomial is $x^2 - 2$. The minimal polynomial of $1/\sqrt{2} = \sqrt{2}/2$ is $2x^2 - 1$ which is not monic, and $1/\sqrt{2}$ is not an algebraic integer.

- (d) If α is an algebraic integer then there is a monic integral polynomial $f(x) = x^n + g(x)$ (with $\deg(f) < n$) such that $f(\alpha) = 0$. The polynomial $h(x) = f(x^2) = x^{2n} + g(x^2)$ is also integral and monic, and $h(\sqrt{\alpha}) = f(\alpha) = 0$. Hence $\sqrt{\alpha}$ is an algebraic integer by question 1. The complex number given in the exercise is an algebraic number, since $\sqrt{2}, \sqrt{3}, 2i$ are algebraic integers, and the square roots and sums of algebraic integers are again algebraic integers. Pari GP suggests the polynomial of degree 8:

$$x^8 + 8x^6 + 64x^4 + 960x^2 + 576. \quad (12)$$

- (e) Write $t = \frac{a}{b}$. Then $z = e^{i\pi t} = e^{\frac{ai\pi}{b}}$ is an algebraic integer since $z^b - 1 = 0$, and similarly $z^{-1} = e^{-\frac{ai\pi}{b}}$. So $2\cos(\pi t) = z + z^{-1}$ is an algebraic integer since it is a sum of two algebraic integers. For $\cos(3\pi/5)$, Pari GP suggests the polynomial $x^2 - x - 1$. This implies that

$$\cos(3\pi/5) = \frac{1 - \sqrt{5}}{4}. \quad (13)$$

(The other root can be eliminated since it is positive but $\cos(3\pi/5)$ is negative)

4. (6 points) Find the gcd of 939 and 753 and find $u, v \in \mathbb{Z}$ such that

$$939u + 753v = \gcd(939, 753). \quad (14)$$

Verify your solution.

Solution: Euclidean division gives that the gcd is 3 and $(-85) \cdot 939 + 106 \cdot 753 = 3$.

5. (10 points) Find a gcd d of $2 + 5i$ and $7 - i$ and find $\alpha, \beta \in \mathbb{Z}[i]$ such that

$$\alpha(2 + 5i) + \beta(7 - i) = d. \quad (15)$$

Verify your solution (using Pari GP for example).

Solution: We use the Gaussian algorithm. First, we divide

$$\frac{2 + 5i}{7 - i} = \frac{(2 + 5i)(7 + i)}{(7 - i)(7 + i)} = \frac{(2 + 5i)(7 + i)}{50} = \frac{9}{50} + \frac{37}{50}i. \quad (16)$$

The closest Gaussian integer is i so

$$\frac{2 + 5i}{7 - i} - i = \frac{9}{50} - \frac{13}{50}i. \quad (17)$$

By multiplying by $7 - i$ we find that the Euclidean division is

$$2 + 5i = i(7 - i) + 1 - 2i \quad (18)$$

where $|1 - 2i|^2 = 1^2 + 2^2 = 5 < |7 - i|^2 = 50$. We repeat this process to find that

$$\frac{7 - i}{1 - 2i} = \frac{(7 - i)(1 + 2i)}{(1 - 2i)(1 + 2i)} = \frac{9}{5} + \frac{13}{5}i. \quad (19)$$

The closest Gaussian integer is $2 + 3i$. We find that

$$\frac{7 - i}{1 - 2i} - (2 + 3i) = -\frac{1}{5} - \frac{12}{5}i \quad (20)$$

so that the Euclidean division is

$$(7 - i) = (2 + 3i)(1 - 2i) - 1. \quad (21)$$

We deduce that $7 - i$ and $1 - 2i$ are coprime, hence $2 + 5i$ and $7 - i$ are coprime by the Euclidean algorithm. The possible gcds are ± 1 and $\pm i$.

To find the linear combination, we take the last division with remainder

$$1 = (2 + 3i)(1 - 2i) - (7 - i). \quad (22)$$

and substitute the previous one

$$1 - 2i = (2 + 5i) - i(7 - i) \quad (23)$$

to get

$$1 = (2 + 3i) [(2 + 5i) - i(7 - i)] - (7 - i) \quad (24)$$

$$= (2 + 3i)(2 + 5i) + (2 - 2i)(7 - i). \quad (25)$$

6. (18 points) The ring of integers of the field $\mathbb{Q}(\sqrt{-3})$ is the ring of Eisenstein integers

$$\mathbb{Z}[\omega] = \{a + b\omega \mid a, b \in \mathbb{Z}\} \quad (26)$$

where $\omega = \frac{1+\sqrt{-3}}{2}$.

- (a) (4 points) What is the monic minimal polynomial of an element $a + b\omega$?
- (b) (4 points) What are the units of $\mathbb{Z}[\omega]$?
- (c) (10 points) Show that $\mathbb{Z}[\omega]$ is a Euclidean domain.

Solution:

- (a) First note that ω is not a rational number, so the minimal polynomial is of degree at least 2. Also, note that for a complex number $z \in \mathbb{C}$, the monic real quadratic polynomial that has z as a root is

$$(x - z)(x - \bar{z}) = x^2 - 2x\Re(z) + |z|^2. \quad (27)$$

Applying this to $z = a + b\omega$ as a root gives the quadratic polynomial

$$x^2 - (2a + b)x + (a^2 + ab + b^2). \quad (28)$$

This polynomial is quadratic, monic, has rational coefficients and $a + b\omega$ as a root. It follows that it is the unique monic minimal polynomial.

- (b) An element $a + b\omega$ is a unit if and only if $a^2 + ab + b^2 = |a + b\omega|^2 = 1$. So we have to find all the integers a, b that satisfy

$$a^2 + ab + b^2 = 1. \quad (29)$$

We can rewrite the LHS

$$a^2 + ab + b^2 = \frac{a^2}{2} + \frac{b^2}{2} + \frac{(a+b)^2}{2}, \quad (30)$$

hence it is a positive integer. First suppose that $|a| \geq 2$ and $|b| \geq 2$. Then we have

$$a^2 + ab + b^2 = \frac{a^2}{2} + \frac{b^2}{2} + \frac{(a+b)^2}{2} \geq \frac{a^2}{2} + \frac{b^2}{2} \geq \frac{2^2}{2} + \frac{2^2}{2} = 4. \quad (31)$$

This shows that we must have $|a| \leq 1$ and $|b| \leq 1$. We see that $a = 0$ and $|b| = 1$ (or $b = 0$ $|a| = 1$) solve the equation $a^2 + ab + b^2 = 1$. In the case where $|a| = |b| = 1$, we can see that $(a, b) = (1, -1)$ and $(a, b) = (-1, 1)$ solve the equations. This gives the 6 units

$$\pm 1, \pm\omega, \pm(1 - \omega). \quad (32)$$

Note that since $\omega^2 - \omega + 1 = 0$, we see that $1 - \omega$ is the inverse of ω .

- (c) The proof is very similar to the proof that $\mathbb{Z}[i]$ is a Euclidean domain.

We take as Euclidean function is $w: \mathbb{Z}[\omega] \rightarrow \mathbb{Z}_{\geq 0}$

$$w(a + b\omega) = |a + b\omega|^2 = a^2 + ab + b^2. \quad (33)$$

It follows from the computations in part *b* that the function is indeed always positive.

Then given nonzero $a + b\omega, c + d\omega$ in $\mathbb{Z}[\omega]$ we write

$$\frac{a + b\omega}{c + d\omega} = \frac{(a + b\omega)(c + d\bar{\omega})}{(c + d\omega)(c + d\bar{\omega})} = p + q\omega \quad (34)$$

for some $p, q \in \mathbb{Q}$, where $\bar{\omega} = \frac{1 - \sqrt{-3}}{2}$ is the complex conjugate of ω . Let $m, n \in \mathbb{Z}$ be closest integers to p, q , so that $|p - m| \leq \frac{1}{2}$ and $|q - n| \leq \frac{1}{2}$. Thus we have

$$\left| \frac{a + b\omega}{c + d\omega} - (m + n\omega) \right|^2 = |(p - m) + (q - n)\omega|^2 \quad (35)$$

$$= (p - m)^2 + (p - m)(q - n) + (q - n)^2 \quad (36)$$

$$\leq |p - m|^2 + |p - m||q - n| + |q - n|^2 \quad (37)$$

$$< \frac{1}{4} + \frac{1}{4} + \frac{1}{4} \quad (38)$$

$$< 1. \quad (39)$$